

Honors Intro Micro | Homework 3

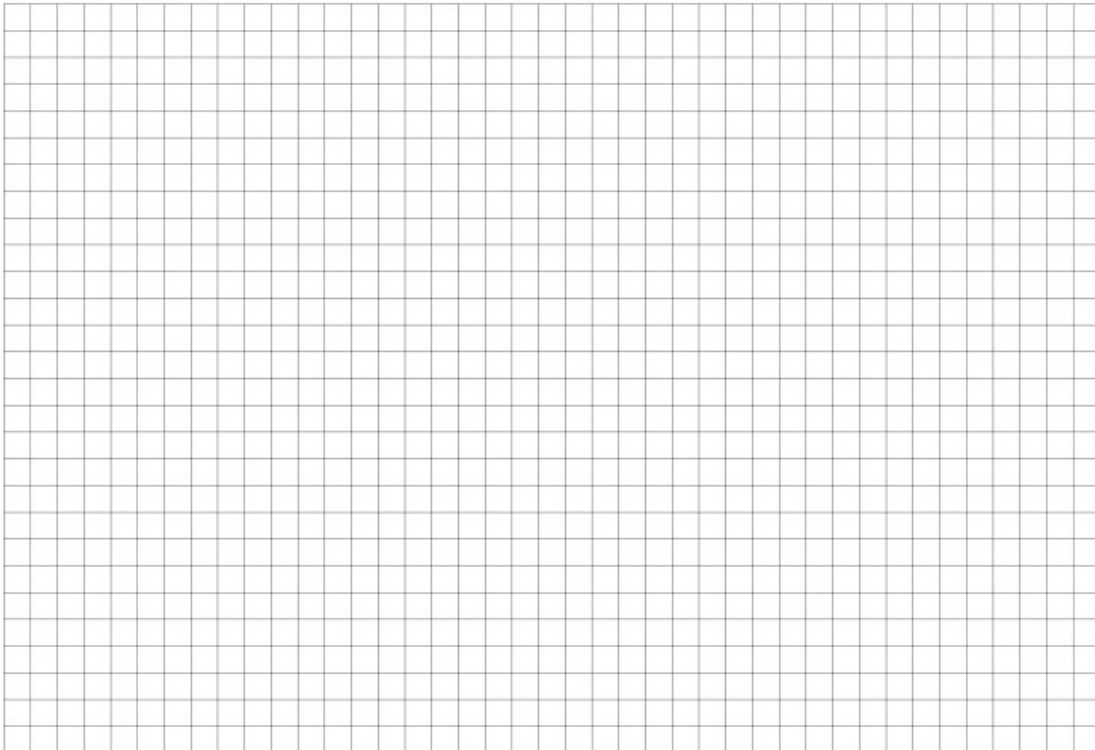
Due Sunday Nov. 7

October 30, 2021

Homework is designed to both test your knowledge and challenge you to apply familiar concepts in new applications. Answer clearly and completely; show your work so I can understand your thought process for partial credit; you are welcomed and encouraged to work in groups as long as your work is your own.

Question 1

Butterbeer is beloved by Hogwarts students and teachers alike and is widely available at wizarding shops. But the beverage has the negative effect of generating displeasing odors from its imbibers. Model the private market for butterbeer using private and social cost and benefit curves. Indicate on your graph the private equilibrium quantity and social equilibrium quantity. Using your economic policy advising credentials, offer a remedy to the malodorous byproduct the Ministry could implement. And conclude with a description of the effects your policy would have on the butterbeer market.



Question 2

Students in Ravenclaw and Hufflepuff share a secondary study room, which is open to all students from both houses at all hours. Every year the heads of house from both houses organize sharing cleaning duties. Cleaning improves the usefulness of the study room for members of both houses, but the chore is inconvenient. If both houses clean, the study room is a fantastic resource. But cleaning takes precious time away from studying, so neither house prefers doing it. The following matrix represents the benefits of each possible strategy combination.

		Hufflepuff	
		Clean	Don't Clean
Ravenclaw	Clean	10, 10	6, 12
	Don't Clean	12, 6	8, 8

Part A

Does Ravenclaw have a dominant strategy? If yes, which strategy is dominant?

Part B

Does Hufflepuff have a dominant strategy? If yes, which strategy is dominant?

Part C

Does this game have a Nash equilibrium? If yes, what is the Nash equilibrium?